

Part A: Conceptual Foundation

1. What are Ensemble Learning Methods and why are they effective?

Ensemble learning methods combine multiple models (base learners) to improve overall performance. They are effective because they reduce errors by averaging predictions, capturing different patterns, and minimizing overfitting compared to a single model.

2. Explain the difference between Bagging, Boosting, and Voting.

Bagging trains multiple models independently on random subsets of data and averages results to reduce variance. Boosting trains models sequentially, focusing on correcting previous errors to reduce bias. Voting combines predictions from different models (same or different types) using majority or weighted voting.

3. What is Bias-Variance Trade-off, and how do ensemble methods address it?

The bias-variance trade-off balances underfitting (high bias) and overfitting (high variance). Bagging reduces variance by averaging multiple models, while boosting reduces bias by improving weak learners. Ensemble methods help achieve better generalization.

4. Explain Voting Classifier and Stacking Ensemble.

A Voting Classifier combines predictions from multiple models using hard (majority) or soft (probability) voting. Stacking uses multiple base models and a meta-learner that learns how to best combine their outputs for improved predictions.

5. Compare AdaBoost, Gradient Boosting, LightGBM, and XGBoost conceptually.

AdaBoost adjusts weights of misclassified samples to focus on errors. Gradient Boosting builds models sequentially by minimizing loss using gradients. XGBoost improves Gradient Boosting with regularization and parallel processing. LightGBM is optimized for speed and efficiency using histogram-based learning and leaf-wise tree growth.